

PHY 456 Tutorial 5:

Review of Spin $\frac{1}{2}$.

Regular wavefunction $\psi: \mathbb{R}^3 \rightarrow \mathbb{C}$

Spinor wavefunction $\psi: \mathbb{R}^3 \rightarrow \mathbb{C}^2$

typically written $\psi(\vec{x}) = \begin{pmatrix} \psi_\uparrow(\vec{x}) \\ \psi_\downarrow(\vec{x}) \end{pmatrix}$.

Then the (differential) operators acting on these spinors ψ become 2×2 matrices.

eg. momentum operator $\hat{p} = \begin{pmatrix} -i\hbar \partial_x & 0 \\ 0 & -i\hbar \partial_x \end{pmatrix}$.

Such matrices must be Hermitian (they're observables).

One can write any Hermitian matrix operator can be written in the "Pauli" basis $\vec{\sigma}$.

ASIDE: Pauli matrices constitute an irreducible representation of the non-relativistic spin group $SU(2)$. (Dirac γ^μ 's: $SL(2, \mathbb{C})$)

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\mathbb{I}_{2 \times 2} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

We also define the 3-tuple

$$\vec{\sigma} = (\sigma_1, \sigma_2, \sigma_3).$$

Immediate consequences:

- $\text{tr}(\sigma_i) = 0.$

- $\det(\sigma_i) = -1.$

- $\sigma_i^2 = \mathbb{I}_{2 \times 2}.$

- $\sigma_i^\dagger = \sigma_i$ (Hermitian)

- $[\sigma_i, \sigma_j] = 2i \epsilon_{ijk} \sigma_k.$

- $\{\sigma_i, \sigma_j\} = 2\delta_{ij} \mathbb{I}_{2 \times 2}$ [Clifford Algebra]

Proving some facts: interested in "spin rotations", that is, rotations "by" elements of $SU(2)$.

Such rotation about an axis $\hat{n}$ act on (spinor) wavefunctions by the unitary operator

$$U(\hat{n}, \alpha) = e^{-i \frac{\alpha \hat{n} \cdot \vec{\sigma}}{2}}$$

$\hat{n}$: rotation axis

$\vec{\sigma}$: pauli "vector"

α : rotation angle.

THEOREM 1:

$$\det(U(\hat{n}, \alpha)) = 1.$$

proof: Since Pauli matrices Hermitian, so is $\hat{n} \cdot \vec{\sigma}$. We can therefore diagonalize it

by a unitary 2x2 matrix V :

$$V^\dagger = V^{-1}$$

$$\hat{n} \cdot \vec{\sigma} = V D V^\dagger, \text{ with } D = \begin{pmatrix} d_1 & 0 \\ 0 & d_2 \end{pmatrix} \quad [d_1, d_2 \in \mathbb{R}].$$

$$\begin{aligned} \text{Then } U(\hat{n}, \alpha) &= e^{-i \frac{\alpha}{2} V D V^\dagger} \\ &= V e^{-i \frac{\alpha}{2} D} V^\dagger. \end{aligned}$$

$$\begin{aligned} \Rightarrow \det(U(\hat{n}, \alpha)) &= \det(V) \det(e^{-i \frac{\alpha}{2} D}) \underbrace{\det(V^\dagger)}_{\frac{1}{\det(V)}} \\ &= \det(e^{-i \frac{\alpha}{2} D}) \\ &= e^{-i \frac{\alpha}{2} (d_1 + d_2)} \\ &= e^{-i \frac{\alpha}{2} \text{tr}(D)} \end{aligned}$$

$$\text{but } \text{tr}(D) = \text{tr}(D V^\dagger V) = \text{tr}(V D V^\dagger) = \text{tr}(\hat{n} \cdot \vec{\sigma}) = 0$$

cyclicality...

$$\text{thus } \det(U(\hat{n}, \alpha)) = e^0 = 1.$$

□

THEOREM 2:

$$U(\hat{n}, \alpha) = \cos\left(\frac{\alpha}{2}\right) \mathbb{I}_{2 \times 2} - i \sin\left(\frac{\alpha}{2}\right) \hat{n} \cdot \vec{\sigma}.$$

Proof: For any matrix A , $e^A = 1 + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$ (Taylor Series)

$$= \sum_{k=0}^{\infty} \frac{A^k}{k!} \quad \text{"functional calculus"}$$

$$\text{Then } e^{-i\frac{\alpha}{2} \hat{n} \cdot \vec{\sigma}} = \sum_{k=0}^{\infty} \frac{(-i\frac{\alpha}{2})^k}{k!} (\hat{n} \cdot \vec{\sigma})^k.$$

$$\begin{aligned} (\hat{n} \cdot \vec{\sigma})^2 &= n_1^2 \sigma_1^2 + n_2^2 \sigma_2^2 + n_3^2 \sigma_3^2 + \cancel{n_1 n_2 (\sigma_1 \sigma_2 + \sigma_2 \sigma_1)} + \cancel{n_1 n_3 (\sigma_1 \sigma_3 + \sigma_3 \sigma_1)} + \cancel{n_2 n_3 (\sigma_2 \sigma_3 + \sigma_3 \sigma_2)} \\ &= (n_1^2 + n_2^2 + n_3^2) \mathbb{I}_{2 \times 2} \\ &= \mathbb{I}_{2 \times 2} \quad \text{since } |\hat{n}| = 1. \end{aligned}$$

By anticommutation.

We can split the $\sum$ into even and odd terms:

$$\text{EVEN } (k = \underline{2p}, p \in \mathbb{Z}^+), \text{ using } (\hat{n} \cdot \vec{\sigma})^k = \mathbb{I}_{2 \times 2}.$$

create identity

$$\text{ODD } (k = \underline{2p+1}, p \in \mathbb{Z}^+), \text{ so } (\hat{n} \cdot \vec{\sigma})^k = (\hat{n} \cdot \vec{\sigma}).$$

$$\text{Then } \sum_{k=0}^{\infty} \frac{(-i\frac{\alpha}{2})^k}{k!} (\hat{n} \cdot \vec{\sigma})^k = \sum_{p=0}^{\infty} \frac{(-i\frac{\alpha}{2})^{2p}}{(2p)!} \mathbb{I}_{2 \times 2} + \sum_{p=0}^{\infty} \frac{(-i\frac{\alpha}{2})^{2p+1}}{(2p+1)!} \hat{n} \cdot \vec{\sigma}.$$

$$\text{Now } \left(-i\frac{\alpha}{2}\right)^{2p} = (-1)^p \left(\frac{\alpha}{2}\right)^{2p}$$

$$\left(-i\frac{\alpha}{2}\right)^{2p+1} = (-i)(-1)^p \left(\frac{\alpha}{2}\right)^{2p+1}$$

$$\text{Thus } \mathcal{U}(\hat{n}, \alpha) = \underbrace{\sum_{p=0}^{\infty} \frac{(-1)^p \left(\frac{\alpha}{2}\right)^{2p}}{(2p)!}}_{\cos(\frac{\alpha}{2})} \mathbb{I}_{2 \times 2} - i \underbrace{\sum_{p=0}^{\infty} \frac{(-1)^p \left(\frac{\alpha}{2}\right)^{2p+1}}{(2p+1)!}}_{\sin(\frac{\alpha}{2})} (\hat{n} \cdot \vec{\sigma})$$

$$e^{-i \frac{\alpha}{2} \hat{n} \cdot \vec{\sigma}} = \cos\left(\frac{\alpha}{2}\right) \mathbb{I}_{2 \times 2} - i \sin\left(\frac{\alpha}{2}\right) (\hat{n} \cdot \vec{\sigma}).$$

